



PART 1: NUMPY (25 QUESTIONS)

1. Create a NumPy array of 5 integers
2. Check data type of array
3. Print first element
4. Print last element
5. Slice elements from index 1 to 4
6. Reverse the array
7. Print alternate elements
8. Add 10 to all elements
9. Subtract 5 from all elements
10. Multiply array by 2
11. Divide array by 2
12. Find square of each element
13. Find sum of array
14. Find mean
15. Find maximum
16. Find minimum
17. Find standard deviation
18. Print elements > 50
19. Print even numbers
20. Replace negative numbers with 0
21. Create 2D array (2x2)
22. Take any 2D Array and its Print its 3rd row and 4th column



PART 2: FOR LOOP

23. Print numbers 1 to 50 which are divisible by 12
24. Print even numbers 1–100 which are divisible by 4 and 6
25. Print table of a number→' 19'
26. Find sum of first n numbers
27. Find factorial
28. Ask user to enter a number and from 1 to that number ,Count numbers which come in that range and are divisible by 3
29. Print reverse numbers (50 to 1)
30. Print square of numbers 1–10
31. Print numbers 1 to 20 by while loop
32. Print odd numbers up to a number which is given input from User by using appropriate look
33. Stop loop when number is 10 (break)
34. Skip multiples of 3 (continue)
35. Ask user to enter 20 times a number if he enters negative numbers then SKIP

36. Restaurant Order System

Problem

“Customer multiple dishes order karta hai

- Har dish ka price enter karo
- Agar price < 0 → skip
- Agar 0 → bill close
- If total > 2000 → give 10% discount”

37. Delivery Tracking System

Problem

'15 Delivery times' record ho rahe hain

Take each delivery's time as a input

IF time < 0 → invalid skip

If time > 60 → late delivery count

38. Library Fine System

Problem

25 Books return ho rahi hain

For every Book, Ask the user "How many Date is Late?"

If 0 → stop

If days > 10 → heavy fine

Else → normal fine

Negative → skip

39. Employee Attendance System

Problem

10 Employees ke 'Work Hours' track ho rahe hain

Ask every employee their hours of work

If hours < 6 → Half Day, also keep a count of half day

If Hours = 9 → Proper Full Day, also keep a count of full day

If hours > 9 → overtime count, keep its count

In the end total employees count should be printed

And the employees doing Half Day, Full Day, Overtime should be counted

Expected Output 📌

Total Employees: 10

Total Employees who took Half Day:8

Total Employees who had Proper Full Day:1

Total Employees who did overtime were: 1

40. Hotel Booking System

Problem

**Ask user for How much Rooms are in booking progress, After getting Input
take the next input of Days**

If days < 0 → skip

If days > 5 → discount apply

If 0 → exit booking

PART 3: DATA STRUCTURES (25 QUESTIONS)

◆ LIST

1. Create a list of numbers
 2. Add element to list
 3. Remove element
 4. Find length
 5. Find max and min
 6. Count even numbers
 7. Reverse list
 8. Sort list
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◆ TUPLE

9. Create a tuple
 10. Access elements
 11. Find length
 12. Convert tuple to list
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◆ SET

13. Create a set
 14. Add element
 15. Remove element
 16. Find union of two sets
 17. Find intersection
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◆ DICTIONARY

18. Create dictionary
 19. Access value using key
 20. Add new key-value pair
 21. Update value
 22. Print all keys and values
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◆ STRING

23. Reverse a string
24. Check palindrome
25. Count vowels in string